

JEE Main - 12 | JEE - 2022

Date: 30/08/2021

Maximum Marks: 300

Timing: 09:00 AM to 02:30 PM

General Instructions

1. The test is of **3 hours** duration.
2. The question paper consists of **3 Parts** (Part I: **Physics**, Part II: **Chemistry**, Part III: **Mathematics**). Each Part has **two** sections (Section 1 & Section 2).
3. **Section 1** contains **20 Multiple Choice Questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE CHOICE** is correct.
4. **Section 2** contains **5 Numerical Value Type Questions**. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.* (Example: 6, 81, 1.50, 3.25, 0.08)
5. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
7. On completion of the test, the candidate must hand over the Answer Sheet to the **Invigilator** on duty in the Room/Hall. **However, the candidates are allowed to take away this Test Booklet with them.**
8. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Marking Scheme

1. **Section – 1:** +4 for correct answer, –1 (negative marking) for incorrect answer, 0 for all other cases.
2. **Section – 2:** +4 for correct answer, 0 for all other cases. There is no negative marking.

Name of the Candidate (In CAPITALS) :

Roll Number :

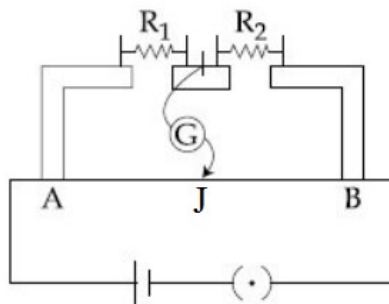
OMR Bar Code Number :

Candidate's Signature : Invigilator's Signature

PART I : PHYSICS**100 MARKS****SECTION-1**

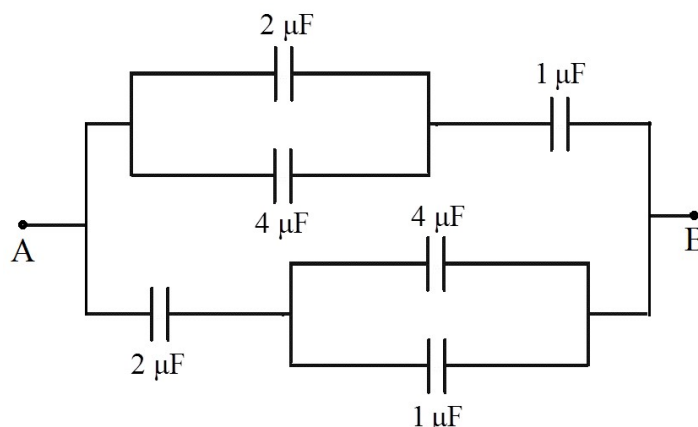
This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

- Consider a metallic spherical shell of inner radius R and outer radius $2R$. A battery is connected between the inner curved surface and the outer curved surface of the shell, such that current flows radially outward. If the electric field at distance $\frac{4R}{3}$ and $\frac{5R}{3}$ from the axis of the shell is E_1 and E_2 respectively, then $\frac{E_1}{E_2}$ is:
 (A) $\frac{5}{4}$ (B) $\frac{4}{5}$ (C) $\frac{25}{16}$ (D) $\frac{16}{25}$
- Two bulbs A and B, rated 200 V, 60 W and 200 V, 40 W respectively are connected in series across an ideal battery of emf E . Each bulb is damaged if the potential difference across it exceeds 250 V. The maximum value of E (in Volts) such that none of the bulbs is damaged is:
 (A) 416.7 (B) 400 (C) 433.3 (D) 450
- A particle of mass m and carrying charge q is moving in a circular path in a plane perpendicular to a uniform magnetic field of intensity B . If the radius of its path is r , the kinetic energy of the particle is:
 (A) $\frac{q^2 r^2}{2mB}$ (B) $\frac{q^2 B^2 r^2}{2m}$ (C) $\frac{qBr^2}{2m}$ (D) $\frac{qr^2}{2mB}$
- In the metre bridge circuit shown, AB is a uniform wire of length 100 cm, and two resistances R_1 and R_2 are connected in the gaps as shown. The current through the galvanometer is zero for length $AJ = 25$ cm. After a resistance $15\ \Omega$ is connected in parallel with R_2 , the current through the galvanometer is zero for length $AJ = 50$ cm. The resistances R_1 and R_2 respectively are:



- (A) 15, 45 (B) 30, 90 (C) 10, 30 (D) 20, 60
- A circular ring of radius R is fixed in the X-Y plane with its centre at the origin and charged uniformly with total charge Q . If the ring is rotated about the Z-axis with constant angular velocity ω , the magnetic field intensity at the point $(0, 0, \sqrt{3}R)$ becomes:
 (A) $\frac{\mu_0 Q \omega}{4\pi R}$ (B) $\frac{\mu_0 Q \omega}{8\pi R}$ (C) $\frac{\mu_0 Q \omega}{16\pi R}$ (D) $\frac{\mu_0 Q \omega}{32\pi R}$
 - A dielectric slab of dielectric constant 4 and thickness $\frac{d}{3}$ is introduced between the plates of an uncharged capacitor with plate separation d . If the capacitor is connected across a battery of emf V , the potential difference across the dielectric slab at steady state becomes:
 (A) $\frac{V}{12}$ (B) $\frac{V}{9}$ (C) $\frac{2V}{15}$ (D) $\frac{V}{6}$

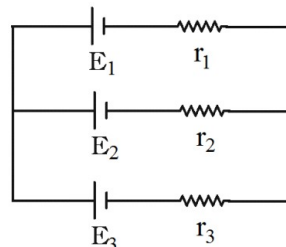
7. A square wire loop of side length a is fixed in the X-Y plane. The loop carries a steady current I in the clockwise direction when viewed from a point on the positive Z-axis. What is magnetic moment of the loop?
- (A) $-Ia^2\hat{k}$ (B) $Ia^2\hat{i}$ (C) $Ia^2\hat{k}$ (D) $Ia^2\hat{j}$
8. A capacitor C is fully charged with voltage V_0 . After disconnecting the voltage source, it is connected in parallel with another uncharged capacitor of capacitance $2C$. The energy loss in the process after the charge is distributed between the two capacitors is:
- (A) $\frac{1}{4}CV_0^2$ (B) $\frac{1}{6}CV_0^2$ (C) $\frac{1}{2}CV_0^2$ (D) $\frac{1}{3}CV_0^2$
9. Consider a circular loop made out of a uniform conducting wire. If an ideal battery is connected between two diametrically opposite points on the loop, the current through the battery is I_0 . Now, if the same battery is connected between two points on the loops separated by a 60° angle, the current through the battery is:
- (A) $\frac{2}{5}I_0$ (B) $\frac{9}{5}I_0$ (C) $\frac{2}{3}I_0$ (D) $\frac{8}{3}I_0$
10. A charged particle is moving in a region with a uniform magnetic field. Consider the following statements:
- (I) at any instant, if the angle between the velocity of the particle and the magnetic field is 60° , the particle must be following a helical path
- (II) the kinetic energy of the particle remains constant
- Which of these options is correct?
- (A) Both (I) and (II) are correct (B) (I) is correct and (II) is incorrect
- (C) (II) is correct and (I) is incorrect (D) Both (I) and (II) are incorrect
11. The equivalent capacitance (in μF) between the points A and B in the following network is:



- (A) $\frac{15}{28}$ (B) $\frac{25}{28}$ (C) $\frac{16}{7}$ (D) $\frac{20}{7}$

12. Three batteries of emf E_1, E_2 and E_3 and internal resistance r_1, r_2 and r_3 are connected as shown. Suppose E_1, E_2, r_1 and r_2 are fixed, and E_3 and r_3 are variable. The current through the battery of emf E_3 can be made zero:

- (A) by changing E_3 or by changing r_3
 (B) by changing E_3 but not by changing r_3
 (C) by changing r_3 but not by changing E_3
 (D) neither by changing E_3 nor by changing r_3

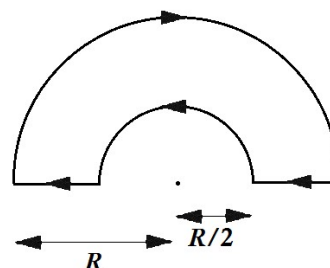


13. A parallel-plate capacitor with vacuum between the plates has capacitance $2 \mu\text{F}$ and is uncharged. A dielectric slab that occupies the entire volume between the plates is introduced inside the capacitor. The dielectric constant of the slab is 4. The capacitor is now charged to a potential difference 100 V. If the capacitor is disconnected from the voltage source, and now the slab is slowly removed from the capacitor, the work done (in milli-Joules) is:

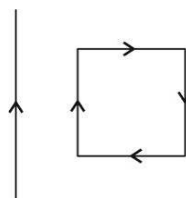
- (A) 30 (B) 40 (C) 120 (D) 200

14. A planar loop consisting of two concentric semi-circular portions of radius R and $\frac{R}{2}$, and two straight portions carries a steady current I as shown. The magnetic field at the common centre of the semi-circular portions is:

- (A) $\frac{\mu_0 I}{4R}$, directed out of the plane of the loop
 (B) $\frac{\mu_0 I}{2R}$, directed out of the plane of the loop
 (C) $\frac{\mu_0 I}{4R}$, directed into the plane of the loop
 (D) $\frac{\mu_0 I}{2R}$, directed into the plane of the loop

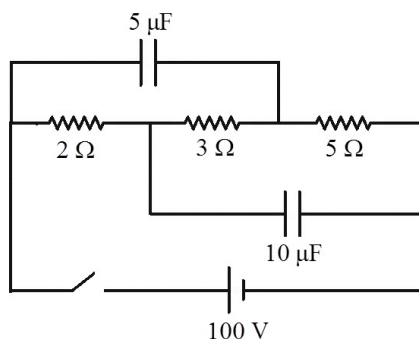


15. A square loop is lying near a long wire in the same plane. The directions of current in the loop and the wire are as shown in figure. Net force on the loop due to this wire is:



- (A) Zero (B) Repulsive (C) Attractive (D) Parallel to the wire

16. In the network shown, the battery is ideal, the switch is initially open and both capacitors are uncharged. The switch is now closed. After a long time, the charge (in μC) on the capacitors of capacitance $5\ \mu\text{F}$ and $10\ \mu\text{F}$ is respectively:



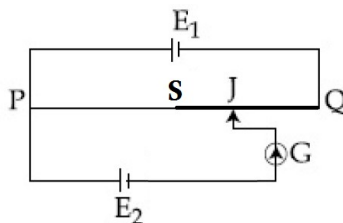
- (A) 250, 500 (B) 500, 1000 (C) 250, 800 (D) 500, 400
17. A small magnetic dipole of magnetic moment m is placed in a uniform magnetic field of intensity B , such that the moment vector makes an angle 30° with the field. The work done in slowly rotating the dipole such that now the moment vector makes an angle 150° with the field is:
- (A) $\frac{1}{2}mB$ (B) mB (C) $\frac{\sqrt{3}}{2}mB$ (D) $\sqrt{3}mB$
18. An infinitely long wire is fixed along the X-axis. The wire carries a steady current in the $-X$ direction. A positively charged particle is moving in the X-Y plane. At a particular instant, the particle is at a point on the positive Y-axis, and has an instantaneous velocity along the unit vector $\left(\frac{-\hat{i} + \hat{j}}{\sqrt{2}}\right)$. The magnetic force on the particle at this instant is along the unit vector:
- (A) $\left(\frac{\hat{i} + \hat{j}}{\sqrt{2}}\right)$ (B) $\left(\frac{\hat{i} - \hat{j}}{\sqrt{2}}\right)$ (C) $\left(\frac{-\hat{i} - \hat{j}}{\sqrt{2}}\right)$ (D) $\left(\frac{-\hat{i} + \hat{j}}{\sqrt{2}}\right)$
19. A capacitor of capacitance C is charged to a potential difference V . After it is disconnected from the voltage source, the capacitor is connected across a resistance R at $t = 0$. The charge that flows through the resistance until $t = 2RC$ is: (e denotes the base of the natural logarithm)
- (A) $\left(1 - \frac{1}{e^2}\right)CV$ (B) $\frac{CV}{e^2}$ (C) $\left(1 - \frac{1}{e}\right)^2 CV$ (D) $\frac{CV}{2e}$
20. A particle of mass 1 g and carrying negative charge of magnitude 0.2 mC is projected horizontally with velocity 500 m/s into a region with a uniform electric field of magnitude 200 V/m directed vertically upwards. Let the $+X$ direction be along the initial velocity of the particle and the $+Z$ direction be vertically upwards. It is desired that the particle passes through the region without deviation (i.e. following a straight line path). To achieve this, there must exist in the region a uniform magnetic field: ($g = 10\text{ m/s}^2$)
- (A) of magnitude 0.3 T along the $+Y$ direction
 (B) of magnitude 0.5 T along the $+Y$ direction
 (C) of magnitude 0.3 T along the $-Y$ direction
 (D) of magnitude 0.5 T along the $-Y$ direction

SPACE FOR ROUGH WORK

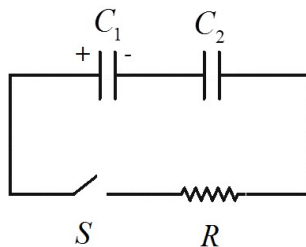
SECTION-2

This Section contains 5 Numerical Value Type Questions. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.* (Example: 6, 81, 1.50, 3.25, 0.08)

21. The balancing length for an unknown cell in a potentiometer experiment is 30 cm. In the balanced condition, the current through the potentiometer wire is 2 A. If the total length of the potentiometer wire is 1.2 m and its total resistance is $40\ \Omega$, the emf of the unknown cell (in Volts) is _____.
22. Two capacitors of capacitance $1\ \mu\text{F}$ and $2\ \mu\text{F}$ are initially uncharged. They are connected in parallel and charged completely using a battery. Let the heat energy dissipated in this charging be H_1 . Now, the capacitors are discharged completely, connected in series and charged completely again using the same battery. Let the heat energy dissipated during this charging be H_2 . The ratio $\frac{H_1}{H_2}$ is _____.
23. In the circuit shown, the potentiometer wire PQ is composed of two parts joined end to end: the part PS has length 60 cm and resistance per unit length $0.1\ \Omega/\text{cm}$, and the part SQ has length 60 cm and resistance per unit length $0.05\ \Omega/\text{cm}$. The main cell E_1 has emf 24 V and internal resistance $3\ \Omega$, and the cell E_2 has emf 15 V. The jockey J is moved on the potentiometer wire PQ to find the balancing point. In the balanced condition (i.e. when the current through the cell E_2 is zero), the length PJ (in cm) is _____.



24. A long cylindrical conductor of radius R carries a current I flowing parallel to its axis. The current density is uniform over the cross-section of the conductor. The magnetic field at a point at a distance $\frac{R}{4}$ from the axis of the conductor is $\frac{\mu_0 I}{n\pi R}$, where n is _____.
25. Two capacitors C_1 and C_2 of capacitance $5\ \mu\text{F}$ and $10\ \mu\text{F}$ respectively are connected with resistance $R = 10\ \Omega$ and a switch as shown. The capacitor C_1 is initially charged to a potential difference 100 V and the capacitor C_2 is initially uncharged. The switch is now closed. At the instant the potential difference across the capacitor C_1 has reduced to 50 V, the current (in Amperes) through the resistance is _____.



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PART II : CHEMISTRY**100 MARKS****SECTION-1**

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

- Consider the following first order competing reactions:

$$X \xrightarrow{k_1} A + B \quad \text{and} \quad Y \xrightarrow{k_2} C + D$$
 if 50% of the reaction of X was completed when 96% of the reaction of Y was completed, the ratio of their rate constants (k_2 / k_1) is: ($\log 5 = 0.7$, $\log 2 = 0.3$)
 (A) 4.06 (B) 0.215 (C) 1.1 (D) 4.65
- A first order reaction is 50% completed in 20 minutes at 27°C and in 5 min at 47°C. The energy of activation of the reaction is: [$\ln 2 = 0.693$]
 (A) 43.85 kJ/mol (B) 55.14 kJ/mol (C) 11.97 kJ/mol (D) 6.65 kJ/mol
- A sample of U^{238} ($t_{1/2} = 4.5 \times 10^9$ yrs) is found to contain 23.8gm U^{238} and 20.6 gm Pb^{206} . The age of ore is: ($\log 2 = 0.3010$)
 (A) 9×10^9 yrs (B) 2.25×10^9 yrs (C) 4.5×10^9 yrs (D) 3.5×10^9 yrs
- For reaction $A \longrightarrow B$, the rate constant $k_1 = A_1 e^{-E_{a1}/RT}$ and for the reaction $P \longrightarrow Q$, the rate constant $k_2 = A_2 e^{-E_{a2}/RT}$. If $A_1 = 10^8$; $A_2 = 10^{10}$ and $E_{a1} = 600$ cal/mol, $E_{a2} = 1200$ cal/mol, then the temperature at which $k_1 = k_2$ is: ($R = 2$ cal/K-mol)
 (A) 600 K (B) 300×4.606 K (C) $\frac{300}{4.606}$ K (D) $\frac{4.606}{600}$ K
- Statement 1 : Time required for 75% radioactive disintegration ($t_{3/4} = 2 \times t_{1/2}$) because
 Statement 2 : Half life ($t_{1/2}$) of the radioactive disintegration is independent of temperature.
 (A) Statement-I is true, statement-2 is true and statement-2 is correct explanation for statement-I.
 (B) Statement-I is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-I.
 (C) Statement-I is true, statement-2 is false.
 (D) Statement-I is false, statement-2 is true.
- The reaction $A(g) \longrightarrow B(g) + 2C(g)$ is a first order reaction with rate constant $3465 \times 10^{-6} s^{-1}$. Starting with 0.1 mole of A in 2 litre vessel, find the concentration of A after 200 sec., when the reaction is allowed to take place at constant pressure and temperature.
 (A) 0.05 M (B) 0.025 M (C) 0.0125 M (D) None of these
- If 0.01% of a substance undergoing decomposition is consumed in 1 millisecond when the concentration is 0.02 M and in 0.25 millisecond when the concentration is 0.04 M, what is the order of the reaction?
 (A) 3 (B) 2 (C) 1 (D) zero
- The dissociation constant of a monobasic acid is 1.6×10^{-5} and the molar conductivity at infinite dilution is $380 \times 10^{-4} \text{ Sm}^2 \text{ mol}^{-1}$. The specific conductance of the 0.01 M acid solution is:
 (A) $1.52 \times 10^{-5} \text{ Sm}^{-1}$ (B) $1.52 \times 10^{-2} \text{ Sm}^{-1}$
 (C) $1.52 \times 10^{-3} \text{ Sm}^{-1}$ (D) None of these

9. Choose the incorrect statement:
- (A) Cell constant values of conductivity cells are independent of the solution filled into the cell
- (B) Kohlrausch law is valid for strong electrolyte but not for weak electrolyte
- (C) In general, conductivity decrease on dilution whereas equivalent and molar conductivity increase on dilution
- (D) Salt bridge is employed to maintain the electrical neutrality and to minimize the liquid-liquid junction potential
10. Select the correct option if it is known that:
 $K_{sp}(\text{AgCl}) > K_{sp}(\text{AgBr}) > K_{sp}(\text{AgI})$
- (A) $E^\circ_{\text{I}^-/\text{AgI}/\text{Ag}} > E^\circ_{\text{Br}^-/\text{AgBr}/\text{Ag}} > E^\circ_{\text{Cl}^-/\text{AgCl}/\text{Ag}}$
- (B) $E^\circ_{\text{I}^-/\text{AgI}/\text{Ag}} < E^\circ_{\text{Br}^-/\text{AgBr}/\text{Ag}} < E^\circ_{\text{Cl}^-/\text{AgCl}/\text{Ag}}$
- (C) $E^\circ_{\text{Ag}/\text{AgI}/\text{I}^-} < E^\circ_{\text{Ag}/\text{AgBr}/\text{Br}^-} < E^\circ_{\text{Ag}/\text{AgCl}/\text{Cl}^-}$
- (D) None of these
11. What is the value of $E_{\text{Cd}^{2+}/\text{Cd}}$ at pH = 10
 Given: $E^\circ_{\text{Cd}/\text{Cd}^{2+}} = 0.40\text{V}$; $K_{sp}(\text{Cd}(\text{OH})_2) = 10^{-14}$
- (A) -0.5773 V (B) -0.227 V (C) $+0.2227\text{ V}$ (D) -0.1045 V
12. 96.5 gm of A oxidized as $\text{A} \longrightarrow \text{A}^{3+} + 3\text{e}^-$, when 2F charge is passed through solution with current efficiency of 90%. What is electrochemical equivalent of A?
- (A) 53.61 (B) 48.25 (C) 5.55×10^{-4} (D) 5.55×10^{-3}
13. Which of the following arrangement will not produce oxygen at anode during electrolysis?
- (A) Dilute H_2SO_4 solution with Cu electrodes
- (B) Dilute H_2SO_4 solution with inert electrodes
- (C) Fused NaOH with inert electrodes
- (D) Dilute NaCl solution with inert electrodes
14. Select correct statement(s):
- (A) The rate constant of a reaction depends on initial concentration of the reactant
- (B) In electrochemical cell, salt bridge is used to mix the solution of anodic & cathodic compartment
- (C) When a lead storage battery is discharged density of solution decreases
- (D) When acetone & trichloro-methane are mixed H-bonding takes place, enthalpy change due to mixing is positive
15. If $E^\circ_{\text{Au}^+/\text{Au}}$ is 1.69 V & $E^\circ_{\text{Au}^{3+}/\text{Au}}$ is 1.40 V, then $E^\circ_{\text{Au}^{3+}/\text{Au}^+}$ will be:
- (A) 0.19 V (B) 2.945 V (C) 1.255 V (D) None of these

16. Surfactant molecules can cluster together as micelles, which are colloid sized cluster of molecules. Micelles form only above critical micelle concentration (CMC) and above certain temperature called Kraft temperature. ΔH of micelle formation can be positive or negative. Which is false about micelle formation?
- (A) ΔS of micelle formation is positive
(B) the hydrophobic part lie towards interior of micelle
(C) the hydrophilic part lie towards surface of micelle
(D) ΔS of micelle formation is negative
17. Give the correct order of initials T (true) or F (false) for following statements
I : Lyophobic sols are irreversible sols.
II : Micelles formation takes place only above kraft temperature
III : PO_4^{3-} ions have more coagulation value than SO_4^{2-} ions for coagulation of positive sols.
IV : The values of the colligative properties observed experimentally are very small of colloidal sols
- (A) FTTF (B) TFTF (C) TTTT (D) TTFT
18. Select the incorrect statement:
- (A) Activation energy is required in physisorption
(B) Extent of physisorption increases as temperature decreases
(C) Extent of adsorption for solute in a solution on solid adsorbent surface depends upon its concentration
(D) Chemisorption is highly specific in nature
19. Which of the following forms a positively charged sol?
- (A) haemoglobin (B) clay (C) eosin (D) charcoal
20. Arrange the decreasing order of coagulation values of electrolytes used for the coagulation of As_2S_3 :
- (A) $\text{NaCl} > \text{BaCl}_2 > \text{MgSO}_4$ (B) $\text{NaCl} > \text{BaCl}_2 = \text{MgSO}_4$
(C) $\text{NaCl} = \text{BaCl}_2 = \text{MgSO}_4$ (D) All are same
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SPACE FOR ROUGH WORK

SECTION-2

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21. Reaction $A + B \longrightarrow C + D$ follows following rate law: $\text{rate} = k[A]^{1/2}[B]^{1/2}$. Starting with initial conc. of one mole of A and B each, what is the time taken (in sec) for amount of A to become 0.25 mole. Given $k = 2.31 \times 10^{-3} \text{ sec}^{-1}$, $\ln 2 = 0.693$
22. A radioactive sample had an initial activity of 56 dpm (disintegration per min). After 69.3 min it was found to have an activity of 28 dpm. Find the number of atoms in a sample having an activity of 10 dpm.
23. At certain temperature, the half-life period in the thermal decomposition of a gaseous substance as follows:
- | | | |
|---------------------|-----|-----|
| P(mmHg) | 500 | 250 |
| $t_{1/2}$ (in min.) | 235 | 950 |
- Find the order of reaction [Given $\log(23.5) = 1.37$; $\log(95) = 1.97$]
24. A solution of 0.1 M CH_3COOH is placed between parallel electrodes of cross-section area 4cm^2 , separated by 2cm. For this solution resistance measured is 100Ω . Calculate elevation in boiling point of the 0.1 M CH_3COOH solution, using following information and assuming molarity equal to molality.
 $K_b = 0.5 \text{ K kg/mol}$; $\wedge_m^\infty(\text{H}^+) = 300 \text{ Scm}^2 \text{ mole}^{-1}$; $\wedge_m^\infty(\text{CH}_3\text{COO}^-) = 100 \text{ Scm}^2 \text{ mole}^{-1}$
 Fill your answer by multiplying it with 800.
25. The molar conductivity of 0.04 M solution of MgCl_2 is $200 \text{ Scm}^2 \text{ mol}^{-1}$ at 298 K. A cell with electrodes that are 4cm^2 in surface area & 8 cm apart is filled with MgCl_2 solution.
 How much current (in Ampere) will flow when the potential difference between the two electrodes is 1000 V?

SPACE FOR ROUGH WORK

PART III : MATHEMATICS**100 MARKS****SECTION-1**

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE CHOICE** is correct.

- $\int_0^{\frac{\pi}{2}} \frac{x \sin 2x}{\sin^4 x + \cos^4 x} dx$ is equal to:

(A) $\frac{\pi^2}{4}$ (B) $\frac{\pi^2}{8}$ (C) $\frac{\pi^2}{16}$ (D) None of these
- The value of $\lim_{n \rightarrow \infty} \left(\frac{1}{n^n} \prod_{r=1}^n (n+r) \right)^{\frac{1}{n}}$ is :

(A) $\frac{1}{e}$ (B) $\frac{2}{e}$ (C) $\frac{4}{e}$ (D) $\frac{8}{e}$
- The area enclosed by $g(x), x = -3, x = 5$ and x -axis where $g(x)$ is the inverse of $f(x) = x^3 + 3x + 1$, is:

(A) $\frac{5}{2}$ (B) $\frac{7}{2}$ (C) $\frac{9}{2}$ (D) $\frac{11}{2}$
- Let $G(x) = \int e^x \left(\int_0^x f(t) dt + f(x) \right) dx$ where $f(x)$ is continuous on R . If $f(0) = 1, G(0) = 0$ then $G'(0)$ equals:

(A) 1 (B) 2 (C) 3 (D) 4
- $\int \frac{\cos^3 x + \cos^5 x}{\sin^2 x + \sin^4 x} dx$ is equal to :

(A) $\sin x - 6 \tan^{-1}(\sin x) + C$ (B) $\sin x - 2 \sin^{-1} x + C$
 (C) $\sin x - 2(\sin x)^{-1} - 6 \tan^{-1}(\sin x) + C$ (D) $\sin x - 2(\sin x)^{-1} + 5 \tan^{-1}(\sin x) + C$

Note: Where C is constant of integration
- The value of the definite integral, $\int_1^{\infty} (e^{x+1} + e^{3-x})^{-1} dx$ is:

(A) $\frac{\pi}{4e^2}$ (B) $\frac{\pi}{4e}$ (C) $\frac{1}{e^2} \left(\frac{\pi}{2} - \tan^{-1} \frac{1}{e} \right)$ (D) $\frac{\pi}{2e^2}$
- $\lim_{n \rightarrow \infty} \sum_{r=1}^{r=4n} \frac{\sqrt{n}}{\sqrt{r} (3\sqrt{r} + 4\sqrt{n})^2}$ is equal to:

(A) $\frac{1}{35}$ (B) $\frac{1}{14}$ (C) $\frac{1}{10}$ (D) $\frac{1}{5}$
- The value of $\lim_{t \rightarrow 0} \ln \left(\frac{1}{t} \int_0^t (1 + 2 \sin 3x)^{4/x} dx \right)$ is equal to :

(A) 6 (B) 12 (C) 18 (D) 24

9. The area bounded by the curve $y = x^2 + 4x + 5$, the axes of co-ordinates & the minimum ordinate is :
- (A) $3\frac{2}{3}$ (B) $4\frac{2}{3}$ (C) $5\frac{2}{3}$ (D) $\frac{8}{3}$
10. The value of definite integral $\int_1^{\sqrt{2}} x \tan^{-1}(x^2 - 1) dx$ equals :
- (A) $\frac{\pi}{4} - \frac{\ln 3}{2}$ (B) $\frac{\pi}{8} - \frac{\ln 2}{4}$ (C) $\frac{\pi}{6} - \frac{\ln 2}{8}$ (D) $\frac{\pi}{2} - \frac{\ln 5}{4}$
11. Let the area enclosed by the curve $y = 1 - x^2$ and the line $y = a$, where $0 \leq a < 1$, be represented by $A(a)$. If $\frac{A(0)}{A(\frac{1}{2})} = k$, then:
- (A) $1 < k < \frac{3}{2}$ (B) $\frac{3}{2} < k < 2$ (C) $2 < k < \frac{5}{2}$ (D) $\frac{5}{2} < k < 3$
12. If $\int (x^9 + x^6 + x^3)(2x^6 + 3x^3 + 6)^{\frac{1}{3}} dx = \frac{1}{A}(2x^9 + 3x^6 + 6x^3)^B + C$, where C is integration constant then AB is equal to:
- (A) 32 (B) 16 (C) 8 (D) 4
13. $\lim_{x \rightarrow \infty} \frac{\int_0^x \tan^{-1} t dt}{\sqrt{x^2 + 1}}$ has the value :
- (A) $\frac{\pi}{2}$ (B) 0 (C) 1 (D) π
14. Consider curves $S_1 : y = \frac{x}{e^x}$, $S_2 : y = x.e^x$, then area bounded by S_1, S_2 and line $x = 1$ is:
- (A) e (B) $2e$ (C) $\frac{1}{e}$ (D) $\frac{2}{e}$
15. If $\int_0^a f(2a - x) dx = 4$ and $\int_0^a f(x) dx = 2$, then $\int_0^{2a} f(x) dx$ is equal to:
- (A) 2 (B) 4 (C) 6 (D) 8
16. The equation $\int_{-\pi/4}^{\pi/4} \left(a|\sin x| + \frac{b \sin x}{1 + \cos x} + c \right) dx = 0$,
Where a, b, c are constants, gives a relation between?
- (A) a, b and c (B) a and c (C) a and b (D) b and c
17. The area of the region bounded by curves $y = e^x, y = e^{-x}, x = 0$ and $x = 1$ is :
- (A) $e + \frac{1}{e}$ (B) $\ln\left(\frac{4}{e}\right)$ (C) $4\ln\left(\frac{4}{e}\right)$ (D) $e + \frac{1}{e} - 2$
18. Let $f : R \rightarrow R$ be a function such that $f(3 - x) = f(3 + x)$ and $f(6 - x) = f(6 + x) \forall x \in R$ and $\int_0^3 f(x) dx = 5$, then the value of $\int_{15}^{45} f(x) dx$ is equal to :
- (A) 25 (B) 30 (C) 50 (D) 60

19. If the primitive of $f(x) = \pi \sin \pi x + 2x - 4$ has the value 3 at $x = 1$, then the number of values(s) of x for which primitive of $f(x)$ vanishes, is:
 (A) 0 (B) 1 (C) 2 (D) more than 2
20. $\int_1^{2013} (x-1)(x-2)(x-3)\dots(x-2013)dx$ equal:
 (A) $(2013)^2$ (B) $(2012)(2013)(2014)$
 (C) $2013!$ (D) 0

SPACE FOR ROUGH WORK

SECTION-2

This Section contains 5 Numerical Value Type Questions. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled. (Example: 6, 81, 1.50, 3.25, 0.08)

21. $\int_{-1}^1 (x - [2x])dx$; equal _____
22. $\int_{\frac{-3\pi}{2}}^{\frac{-\pi}{2}} [(x + \pi)^3 + \cos^2(x + 3\pi)]dx$ is _____
23. The value of the integral $\int_0^{\frac{1}{2}} \frac{1 + \sqrt{3}}{((x+1)^2(1-x)^6)^{\frac{1}{4}}} dx$ is _____
24. If $\alpha = \int_0^1 (e^{9x+3\tan^{-1}x}) \left(\frac{12+9x^2}{1+x^2} \right) dx$ where $\tan^{-1}x$ takes only principal values, then the value of $\left(\log_e |1 + \alpha| - \frac{3\pi}{4} \right)$ _____
25. If $\int_0^\pi \frac{x^2 \sin 2x \sin \left\{ \frac{\pi}{2} \cos x \right\}}{2x - \pi} dx = \frac{A}{\pi}$, then A is _____

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End of JEE Main - 12 | JEE 2022